DOUBLE JEOPARDY: The Vulnerability Spiral

Sea-Level Exposure and Reef Heat Stress Across 5 Island Groups

Executive Summary · Version 2.1 · DOI: 10.5281/zenodo.23827474 · Sakshi D. Maske

Project Overview

Low islands face two threats at once: the sea rising, and the reefs and mangroves that hold it back weakening. The project measures both across 5 island groups in 3 ocean basins (Maldives, Lakshadweep, Seychelles, Fiji and the Canary Islands), and checks whether protection, and the rules on where people may build, follow the risk.

Version 1 reported Seychelles as most exposed, significant coral trends, a ranking robust to weighting and protection tracking risk. Rechecking each measurement choice withdrew most of those results. Version 2 shows how much the answer depends on choices that are often not reported: which elevation model, what is counted as exposure, and how the inputs are weighted.

The Question

How many people on these islands live at or below 1 m? How has heat stress on their reefs changed between 1996 and 2020, and have mangroves declined? When the two are put side by side, which island comes out highest, and does protected-area coverage follow that ranking? Finally, do the governments' coastal setbacks reach the buildings that stand on low land, and does having more coastal rules make a difference?

The Method

- **Exposure:** share of people (WorldPop 2020, 100 m) on land at or below 1 m, on the **DeltaDTM v1.1 bare-earth model** (Pronk et al. 2024). Copernicus GLO-30, a surface model, is kept for comparison. The share of OpenStreetMap named places is a second measure; 0.5 m and 1.5 m are sensitivity checks. Heights are above the EGM2008 geoid, used as an approximation of mean sea level.
- **Reef heat stress:** NOAA Coral Reef Watch Degree Heating Weeks (DHW, 5 km), averaged over all mapped reef cells per island, 1996–2020. Change between 1996–2000 and 2016–2020, and a seasonal Mann-Kendall test on monthly means corrected for serial correlation.
- **Mangroves:** Global Mangrove Watch v3 at 1996, 2010 and 2020.
- **Compound score:** exposure and change in reef heat stress, each scaled 0–1, averaged; weighting swept from 0 to 100%.
- **Governance:** share of the 10 km coastal zone inside a protected area (WDPA, June 2026, overlaps merged). Descriptive only.
- **Coastal setbacks:** about 1.35 million building footprints (Overture Maps, 2026-08-19.0), each with its bare-earth elevation and distance to the shoreline, compared with the setback in national law: 100 m in the Canary Islands (Ley de Costas), 25 m in Seychelles (S.I. 91 of 2023), 20 m in Lakshadweep (ICRZ 2019), none for inhabited land in the Maldives and Fiji. Instruments are classified with Hood's NATO scheme, and the Maldives–Lakshadweep pair is read as a most similar systems comparison.
- **Policy coding:** each government's coastal rules in force in October 2026 coded by hand from legal texts: policy density (six instrument types) and setback intensity (width), following Knill, Schulze and

Tosun (2012), plus 23 dated rule changes since 1988 coded as expansion, dismantling or a shift between levels of government. Every value carries its source.

The Findings

People at or below 1 m

Island group	Bare earth (DeltaDTM)	Surface (Copernicus)	People
Maldives	11.82%	8.88%	57,889
Lakshadweep	6.40%	0.79%	4,238
Seychelles	2.07%	0.43%	1,933
Fiji	0.28%	0.21%	2,460
Canary Islands	0.19%	0.15%	3,946

Hypothesis	Result
H1: Reef heat stress is rising	**Partly supported.** Mean DHW over reef rose on every reef island (Seychelles +0.52, Fiji +0.29, Maldives +0.07, Lakshadweep +0.05 °C-weeks), but no monotonic trend is significant once serial correlation is accounted for ($p = 0.14$ to 0.90)
H2: Mangroves are declining	**Not supported.** No decline on any of the 3 islands (Fiji 485.7 → 488.4 km^2)
H3: Protection follows risk	**Not supported.** $r = -0.10$ (95% interval −0.97 to 0.95, $n = 4$); coverage from 1.3% (Maldives) to 71.4% (Seychelles)

Coastal setbacks against low-lying buildings

Island group	Buildings at or below 1 m	Statutory setback	Low buildings landward of every setback	Distance covering 90% of them
Maldives	13.4% (15,392)	none for inhabited islands	99.4%	225 m
Lakshadweep	4.7% (444)	20 m	93.2%	345 m
Seychelles	4.6% (2,440)	25 m	92.0%	427 m
Fiji	0.5% (1,397)	none	99.8%	463 m
Canary Islands	0.02% (192)	100 m (20 m urban)	15.1%	122 m

On the four tropical island groups the setbacks reach fewer than one in ten low buildings, which stand 80–136 m from the shore (median). The Canary Islands' 100 m line covers 85%, because there low land is a thin beachfront strip. Even a 100 m rule would cover only 54% of the low buildings in the Maldives, while putting 39% of all buildings inside it: on atolls, low land is the island rather than its edge.

Policy outputs against their reach

	Canary Islands	Lakshadweep	Seychelles	Fiji	Maldives
Coastal instrument types in force (of 6)	6	5	3	3	2
Low buildings reached by the setback	84.9%	6.8%	8.0%	0%	0%

More rules did not mean more reach on the tropical islands: Lakshadweep has five of six instrument types and reaches 7% of its low buildings. Of 23 rule changes since 1988, 15 added protection; Spain relaxed its coastal law in 2013, and the Maldives widened exemptions from environmental review in 2024–2025. Regime type (V-Dem) does not sort these changes: rules were added under scores from 0.13 to 0.89, and relaxed at 0.88 (Spain) and 0.56–0.58 (Maldives).

Compound score (equal weights): Seychelles 0.58 · Maldives 0.56 · Lakshadweep 0.32 · Fiji 0.28 · Canary Islands 0.00. Exposure points to the Maldives and reef heat stress to Seychelles, so the top island switches to the Maldives once exposure carries 52% or more of the weight. The top two are a near tie.

What Changed From Version 1, and Why

- ✗ **"78.3% of Seychelles settlements at risk."** 188 of 244 Seychelles points read the elevation raster's NoData value (−32767 m) and were counted as at risk. And most OSM "settlement" points on these islands are names of islands and islets, some sitting in open water. I now count people, with named places as a second measure (Seychelles 24.7%, Maldives 28.8% on bare earth).
- ✗ **Surface elevation model.** Copernicus measures roofs and canopy on wooded atolls. On bare earth, Lakshadweep goes from 0.8% to 6.4% of people at or below 1 m. Where Copernicus is above 30 m, I do not use DeltaDTM: on small forested granite islands in Seychelles it otherwise reads a hill as sea level.
- ✗ **"Coral trends significant for Maldives (p = 0.0046) and Seychelles (p = 0.0069)."** That used one 5 km cell per island and a test that treats neighbouring months as independent. DHW is a 12-week running sum, so they are not. Without the correction, Seychelles (p = 0.002) and the Maldives (p = 0.049) still look significant; with it, p = 0.14 and 0.31.
- ✗ **"The ranking holds at every weighting."** Seychelles came out top at every weighting because it was highest on both inputs by construction. With people and reef-cell means, the inputs disagree and the ranking depends on the weights.
- ✗ **"Governance correlation r = 0.965, p = 0.035."** Overlapping protected areas were counted twice and coverage was divided by land area. As a share of the coastal zone, the relationship disappears.
- ✗ **Seychelles built-up rise (+0.11 NDBI).** Measured over bounding boxes that were mostly ocean. On land only, Seychelles shows no rise and the Maldives a small one (+0.055).

Limitations

- These are elevation thresholds, not flood models: no tides, surge, waves or connectivity. DeltaDTM's error against airborne lidar (about 0.4–0.5 m) is large next to a 1 m threshold, so the comparison between models matters more than any single percentage.
- WorldPop is a modelled population surface, not a census.
- Five island groups (four for governance) describe this sample; they cannot establish general relationships.

- The compound score uses min–max scaling, so Canary Islands' 0.00 is the lowest in this sample, not an absence of risk, and it leaves out adaptive capacity entirely.
- Mangrove patches on the atolls are close to Global Mangrove Watch's pixel size, so small losses there may not be detectable.
- An independent recalculation of the coral trend test and of the setback distances (on the WGS84 ellipsoid) reproduces the reported results.
- Setback distances are measured to the island polygon edge, not the legal high water mark, and only national law is compared; local plans and enforcement are not observed. The Lakshadweep width rests on secondary sources, Fiji's buildings are OpenStreetMap only, and 45% of the low Seychelles buildings sit on DeltaDTM cells below 0 m (excluding them changes the result by under three points).

Real World Relevance

Adaptation finance for small islands is increasingly tied to vulnerability indices. This project shows that the same islands can rank differently depending on the elevation model, the exposure measure and the weighting, and that a surface model can understate how many people live at or below 1 m by several times on a forested atoll. Any index built on these inputs should report those choices and how sensitive its ranking is to them. For adaptation law, the setback result suggests that on atolls the rules on where to build should be written in heights (minimum floor or ground levels) rather than in distance from the shore. A two-page policy brief sets this out for practitioners (`paper/DJ_Policy_Brief.md`).

GitHub: github.com/sakshimaske303-commits/DOUBLE-JEOPARDY | Live Dashboard: double-jeopardy-6ev9trz3dwafsb7panbnxg.streamlit.app | Zenodo DOI: 10.5281/zenodo.23827474

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